

2D Image Recognition Using SIFT

Computer Vision: CS 566

Computer Science

University of Wisconsin-Madison

2D Recognition Using SIFT

Recognize 2D objects in cluttered scenes using the Scale Invariant Feature Transform (SIFT).

Topics:

- (1) Local Appearance and Interest Points
- (2) Blob Detection
- (3) Scale-Space
- (4) The SIFT Interest Point Detector
- (5) Matching SIFT Descriptors

A Little Quiz

How would you recognize the following types of objects?



Template



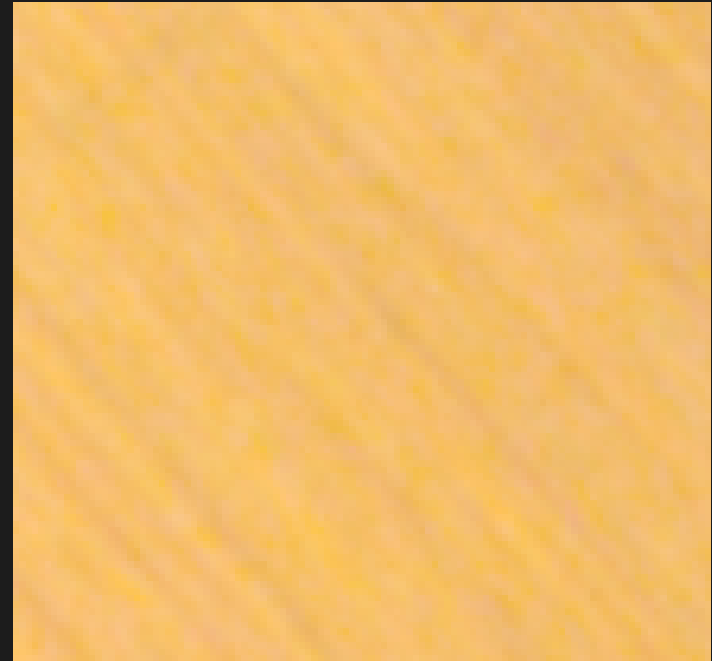
Rich 2D image

Match “Interesting Points or Features”

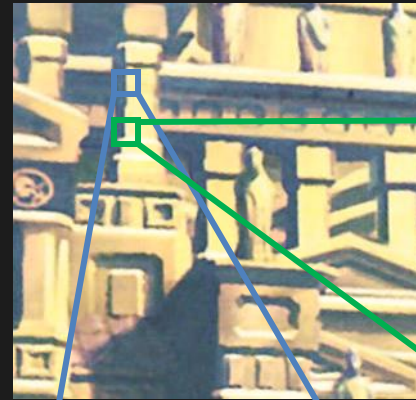
What is an Interesting Point/Feature?

- Has **rich image content** (color variations, gradient variations, ...) within the local window
- Has well-defined **representation (signature)** for matching/comparing with other points
- Has a well defined **position** in the image
- Should be **invariant to image rotation** and **scaling**
- Should be relatively **invariant to lighting** changes

Some Patches are not “Interesting”

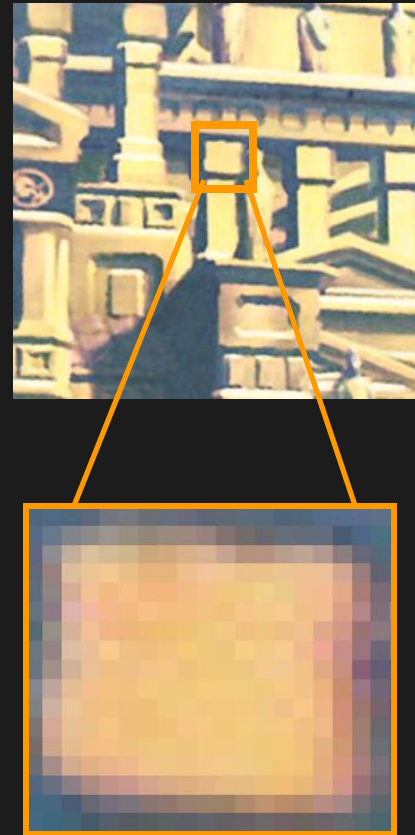
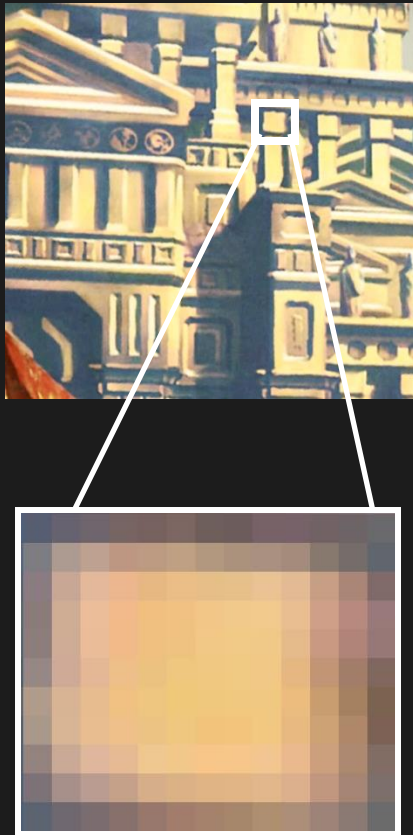


Are Lines/Edges Interesting?



Cannot “**Localize**” an Edge

Are Blobs Interesting?



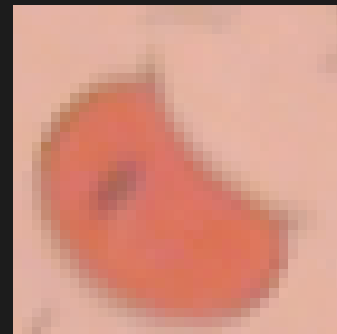
Yes! Blobs have **fixed position** and definite **size**.

Raw Images are Hard to Match



Different size, orientation, lighting, brightness, ...

Removing Sources of Variation



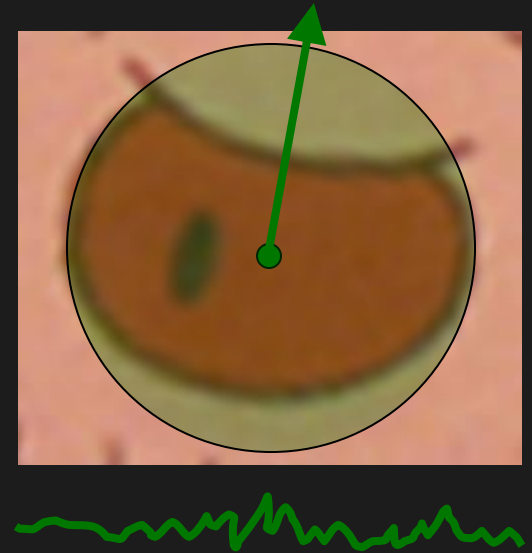
Matching becomes easier if we can
remove variations like size and orientation.

Blobs as Interest Points

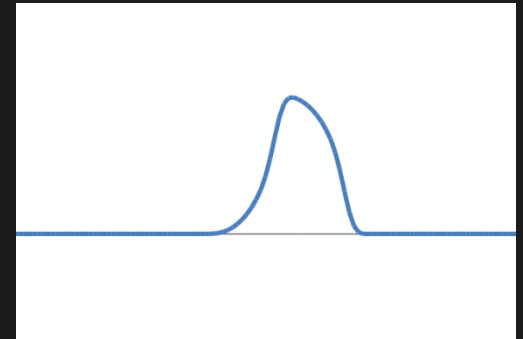
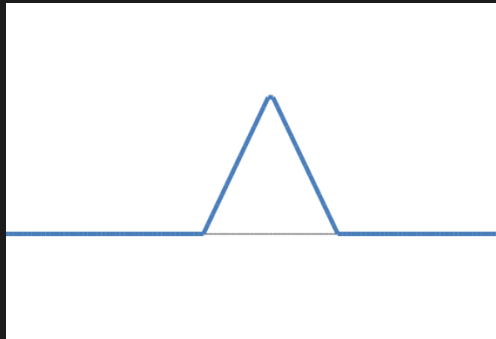
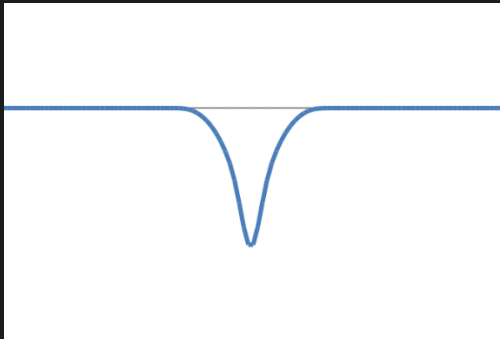
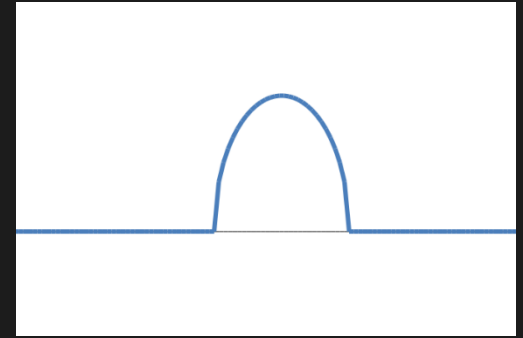
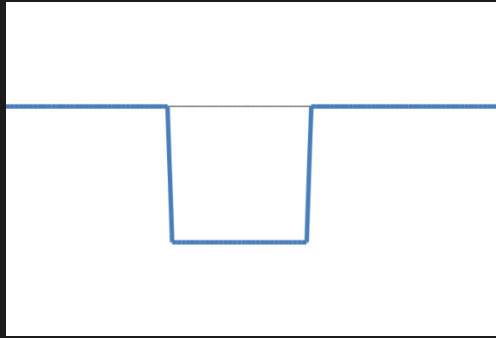
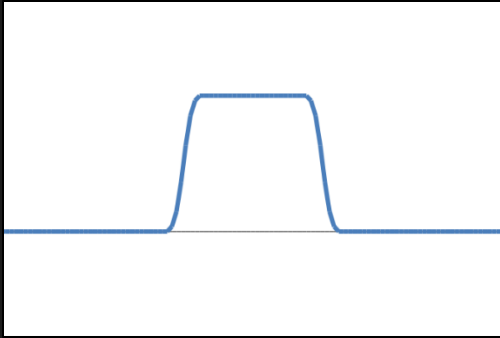
We will use **Blob-like Features** for 2D recognition.

We need to:

- **Locate** a blob
- Determine its **size**
- Determine its **orientation**
- Formulate a **description** or signature that is independent of size and orientation

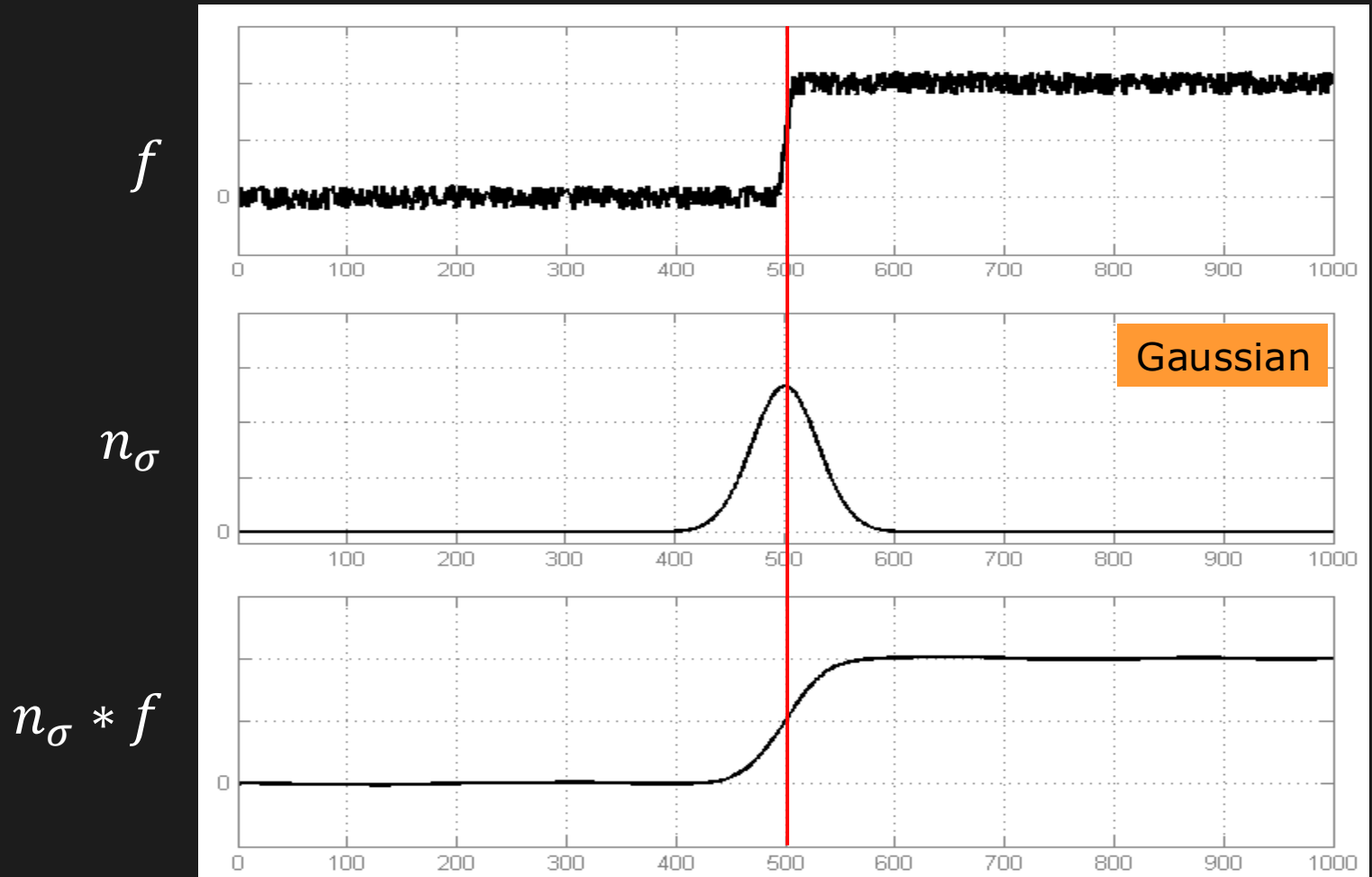


1D Blobs



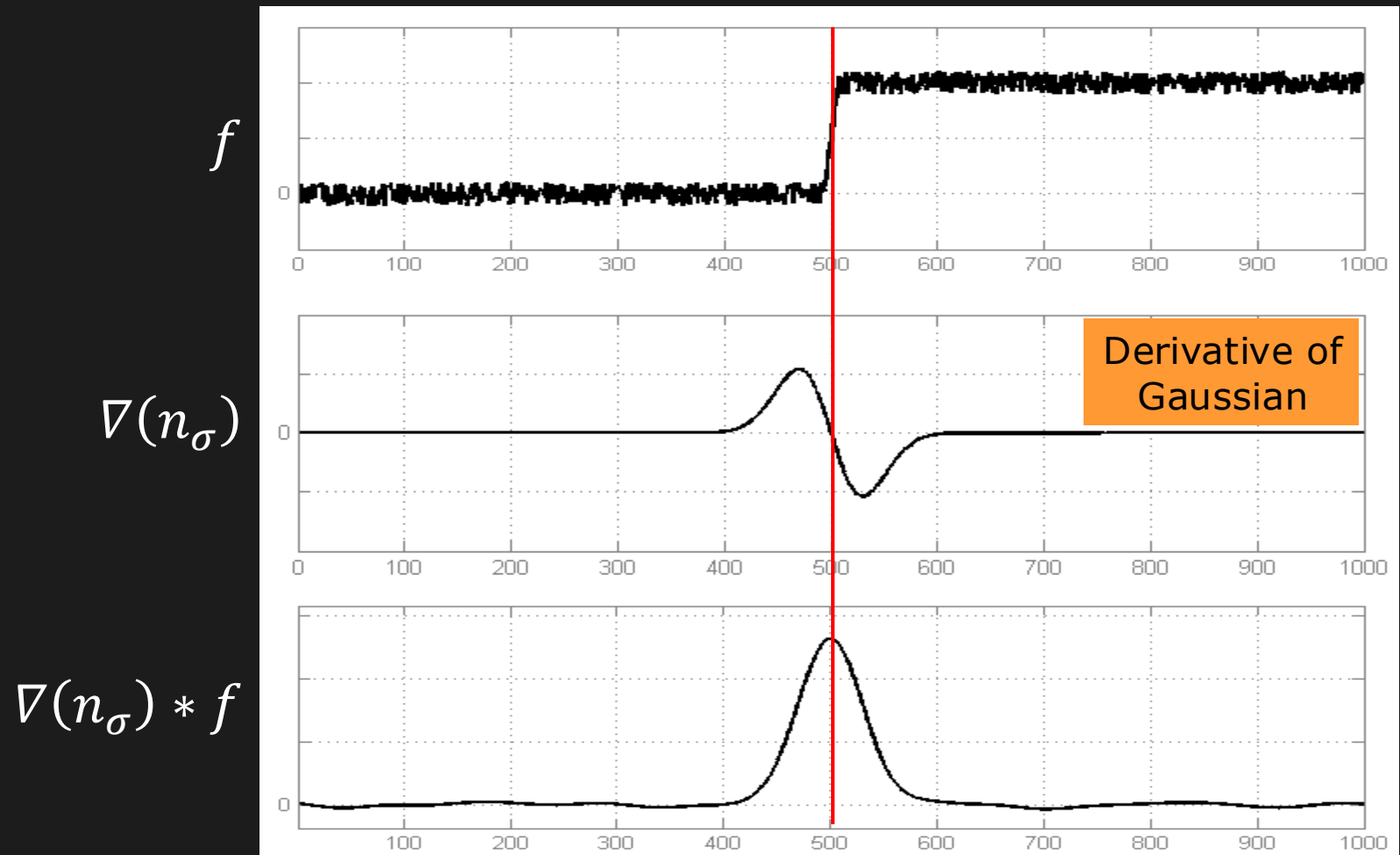
Examples of 1D Blob-like structures

Review: Gaussian Filter



Gaussian Filter is used for removing noise by smoothing

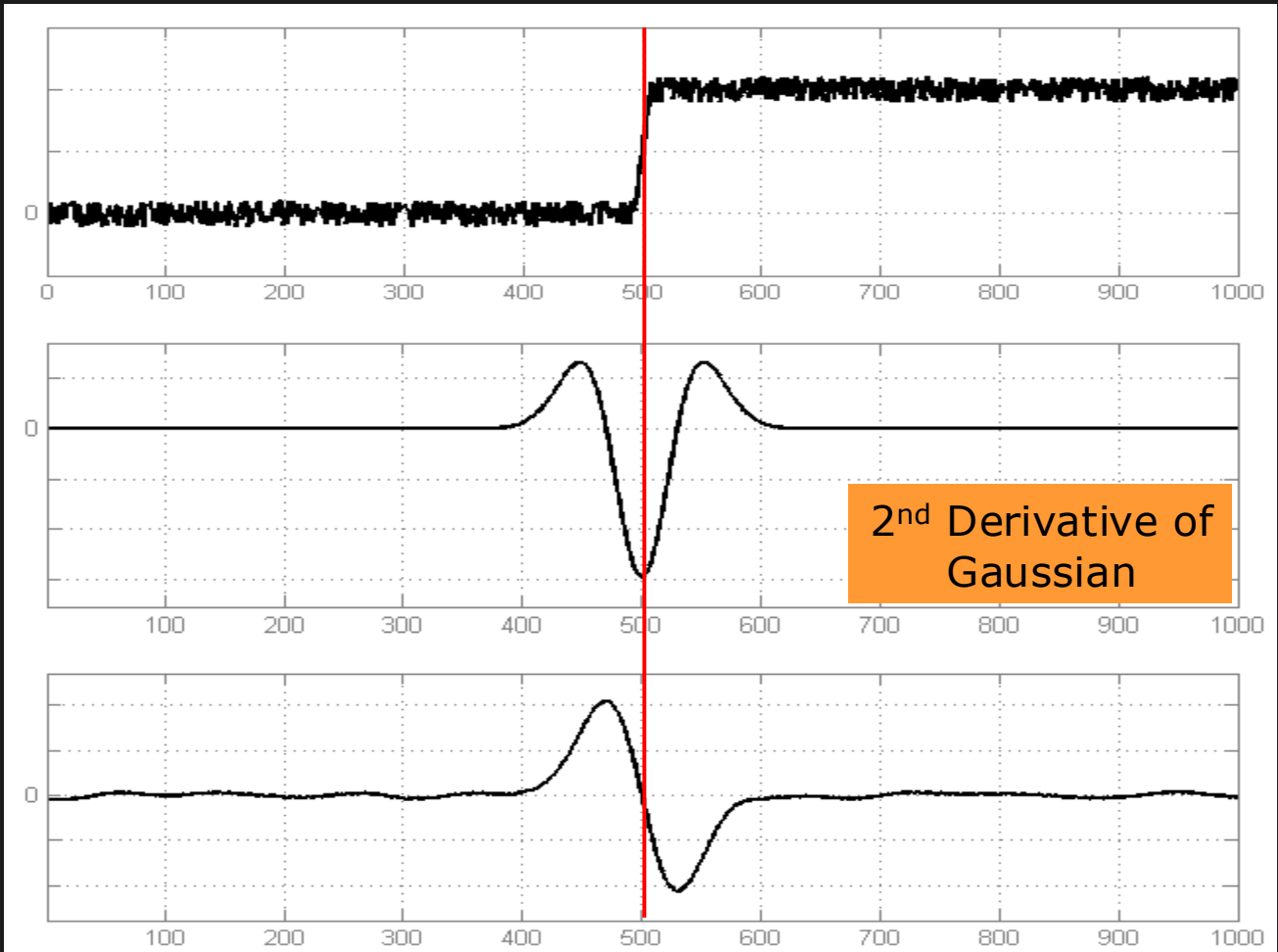
Review: Derivative of Gaussian



Extremum of Derivative of Gaussian denotes an Edge

Review: 2nd Derivative of Gaussian

f

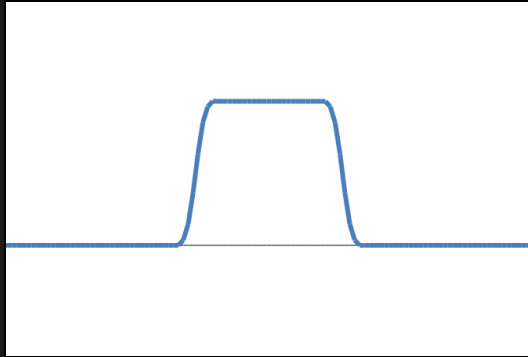


$\nabla^2(n_\sigma)$

$\nabla^2(n_\sigma) * f$

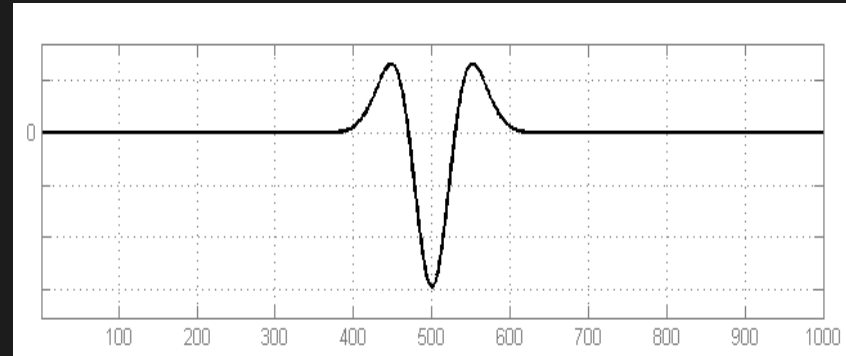
Zero Crossings in 2nd Derivative of Gaussian denote an Edge

Laplacian Filter on 1D Blobs

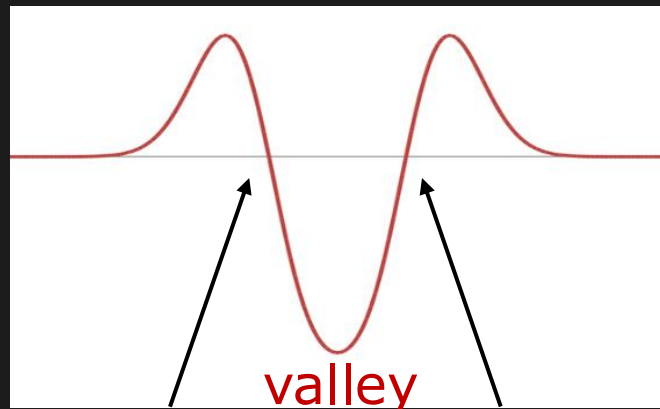


1D Blob

*

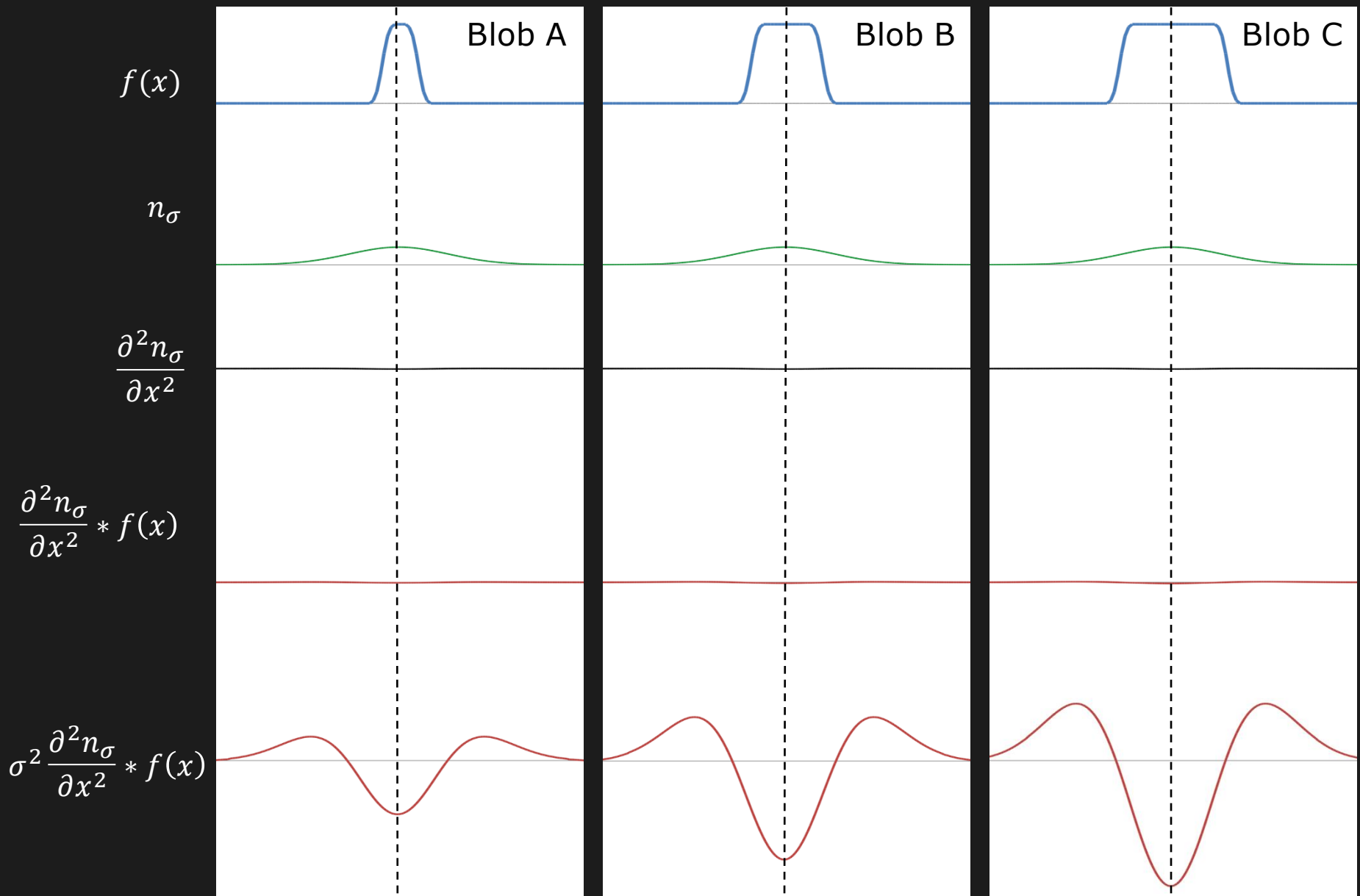


Laplacian of Gaussian

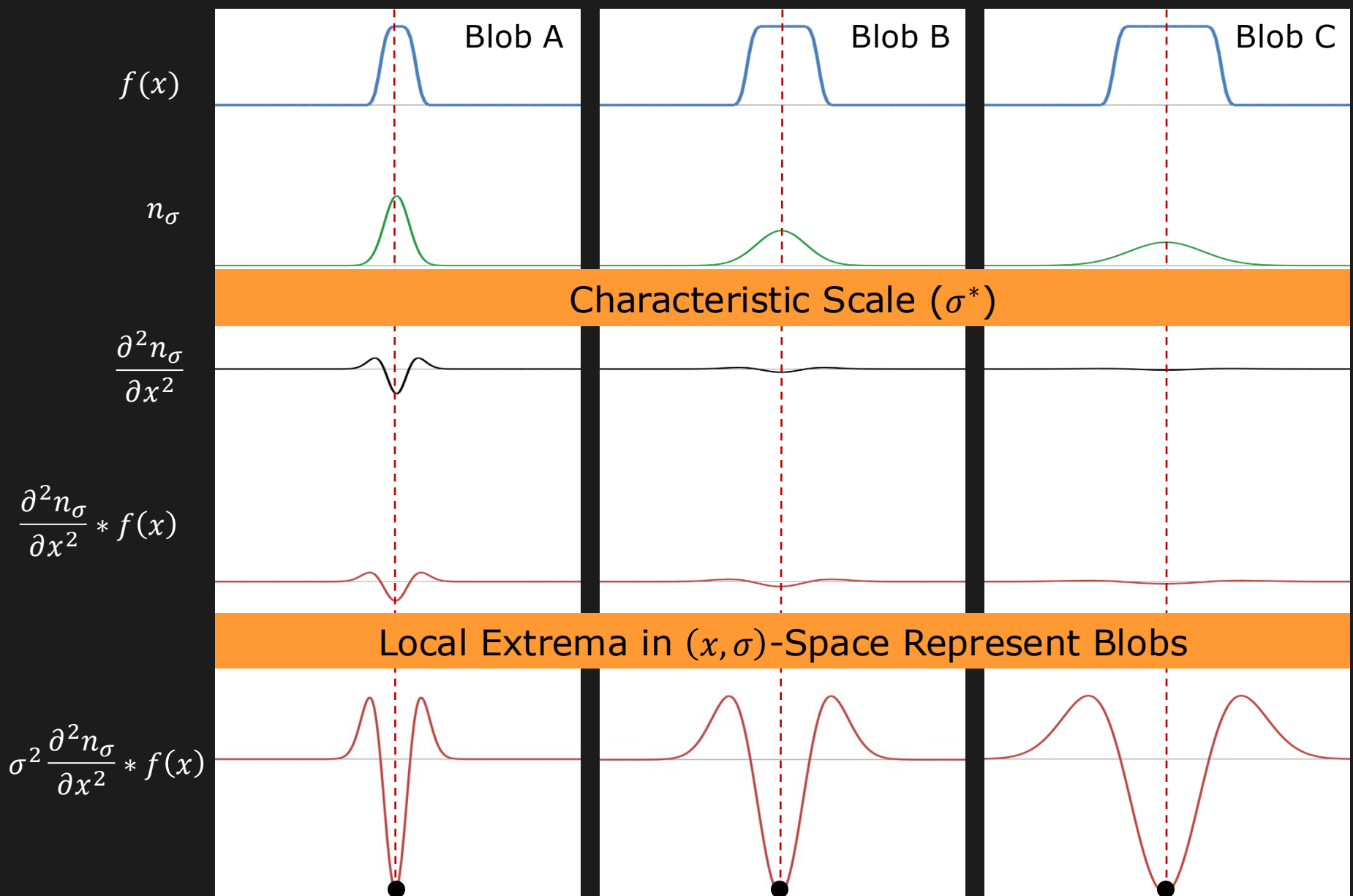


two consecutive zero crossings

1D Blob and 2nd Derivative of Gaussian



1D Blob and 2nd Derivative of Gaussian



1D Blob Detection Summary

Given a 1D signal $f(x)$.

Compute $\sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x)$ at many scales $(\sigma_0, \sigma_1, \sigma_2, \dots, \sigma_k)$.

Find:
$$(x^*, \sigma^*) = \arg \max_{(x, \sigma)} \left| \sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x) \right|$$

x^* : Blob Position

σ^* : Characteristic Scale (Blob Size)

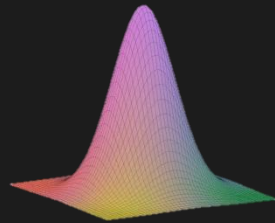
2D Blob Detector

Normalized Laplacian of Gaussian (NLoG) is used as the 2D equivalent for Blob Detection.

Laplacian

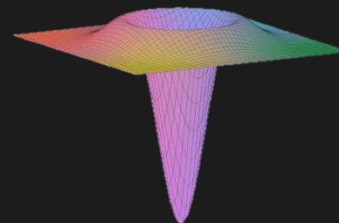
$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

Gaussian



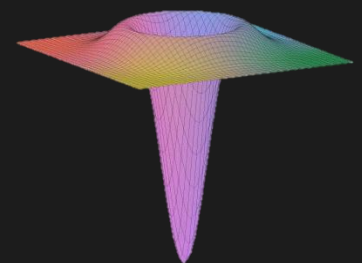
n_σ

LoG



$\nabla^2 n_\sigma$

NLoG

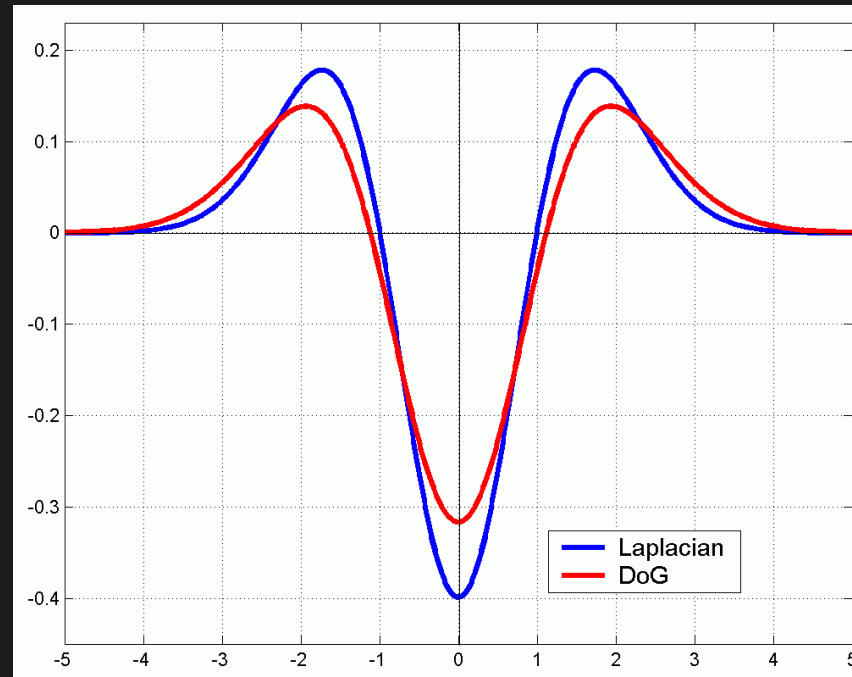


$\sigma^2 \nabla^2 n_\sigma$

Location of Blobs given by **Local Extrema** after applying Normalized Laplacian of Gaussian at many scales.

Fast NLoG Approximation: DoG

$$\text{Difference of Gaussian (DoG)} = (n_{s\sigma} - n_{\sigma}) \approx (s - 1)\sigma^2 \nabla^2 n_{\sigma}$$

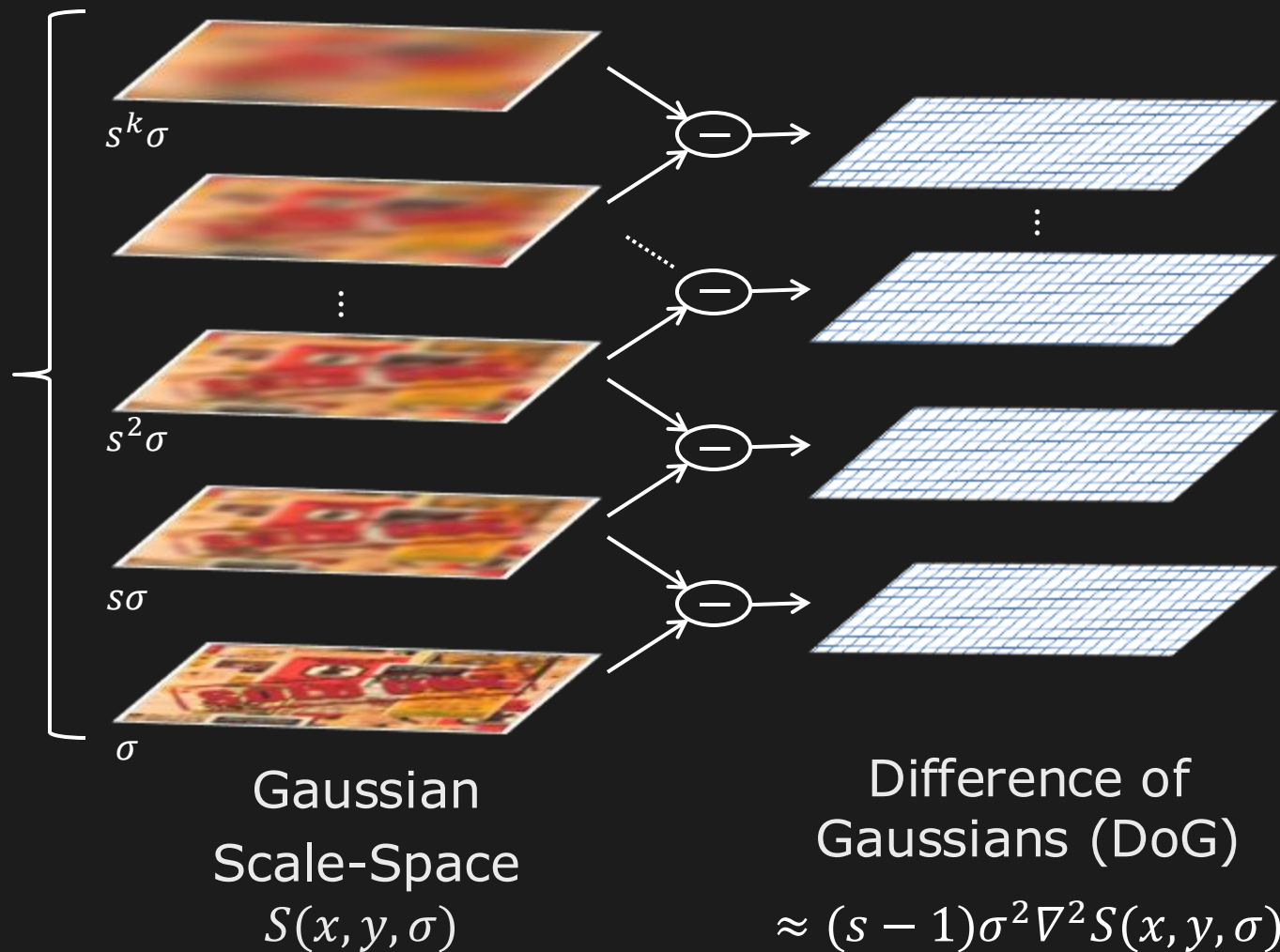


$$\text{DoG} \approx (s - 1) \text{NLoG}$$

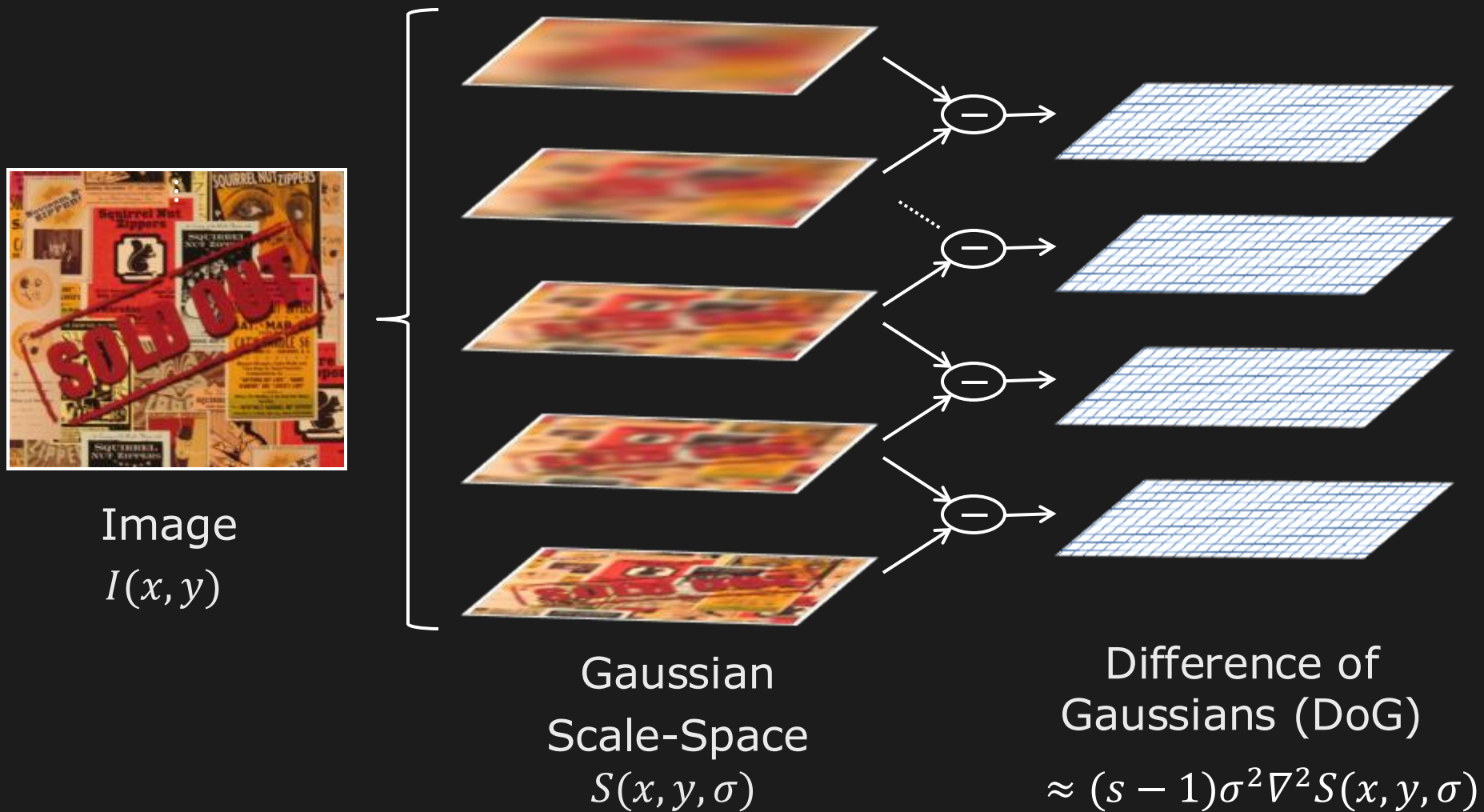
Extracting SIFT Interest Points



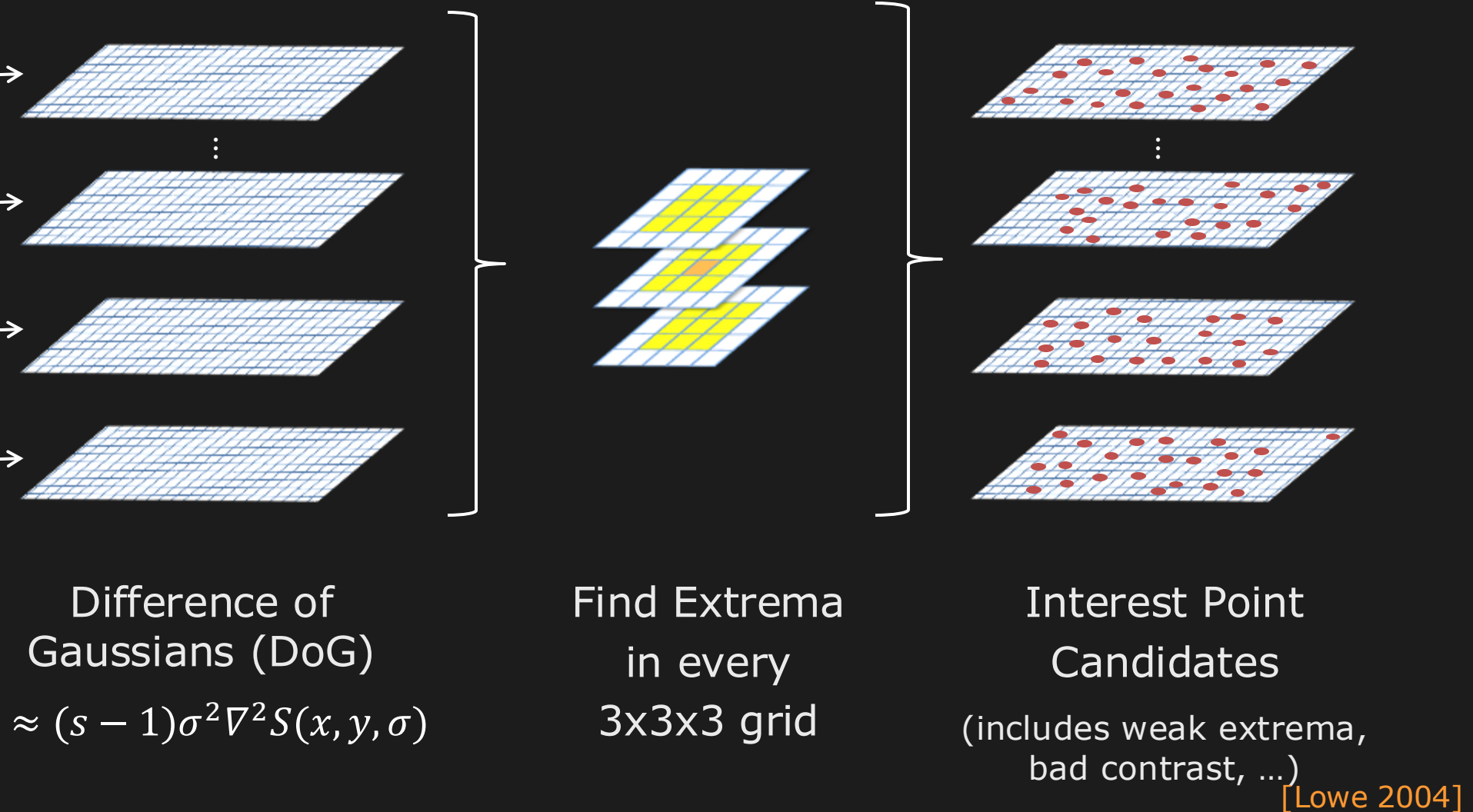
Image
 $I(x, y)$



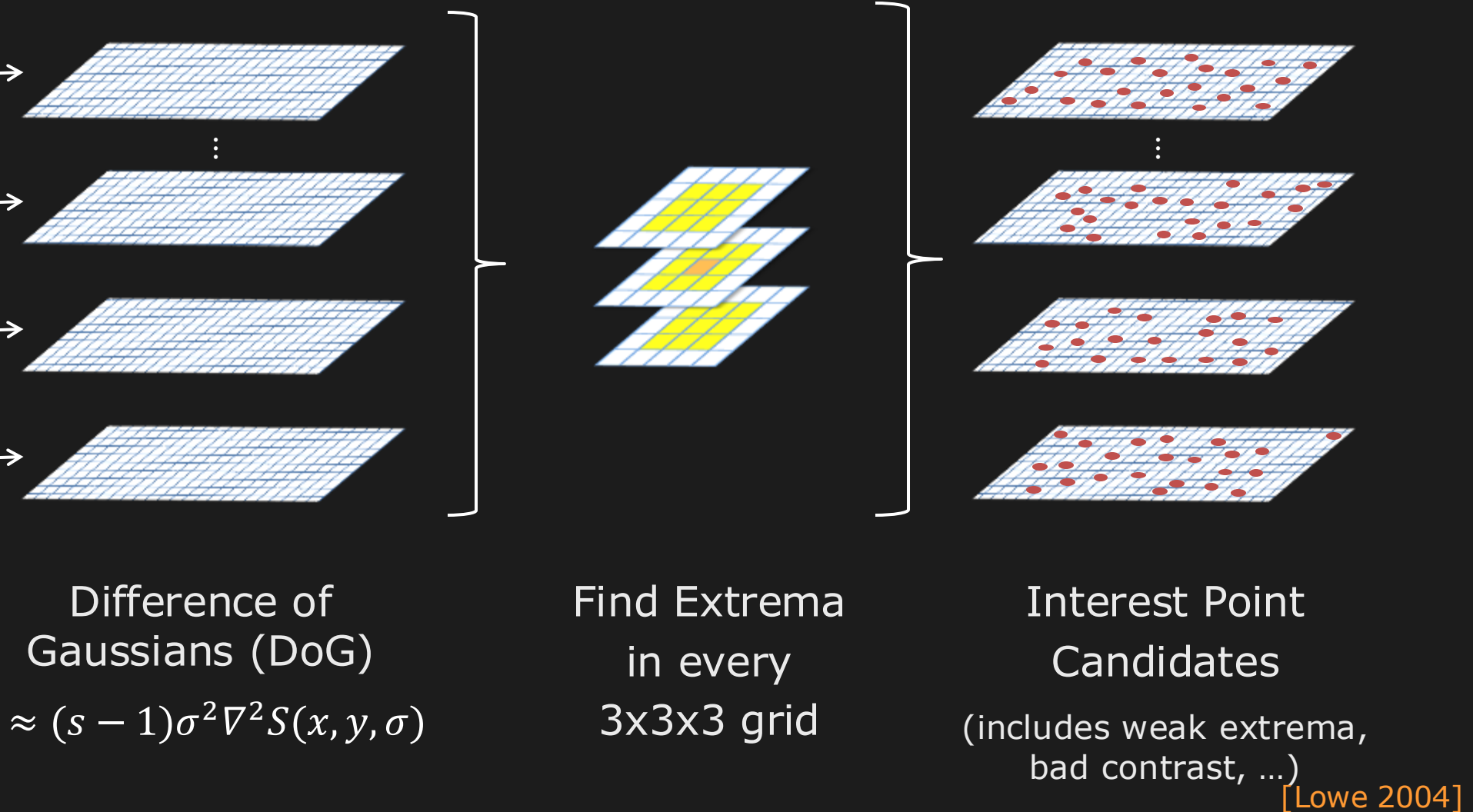
Extracting SIFT Interest Points



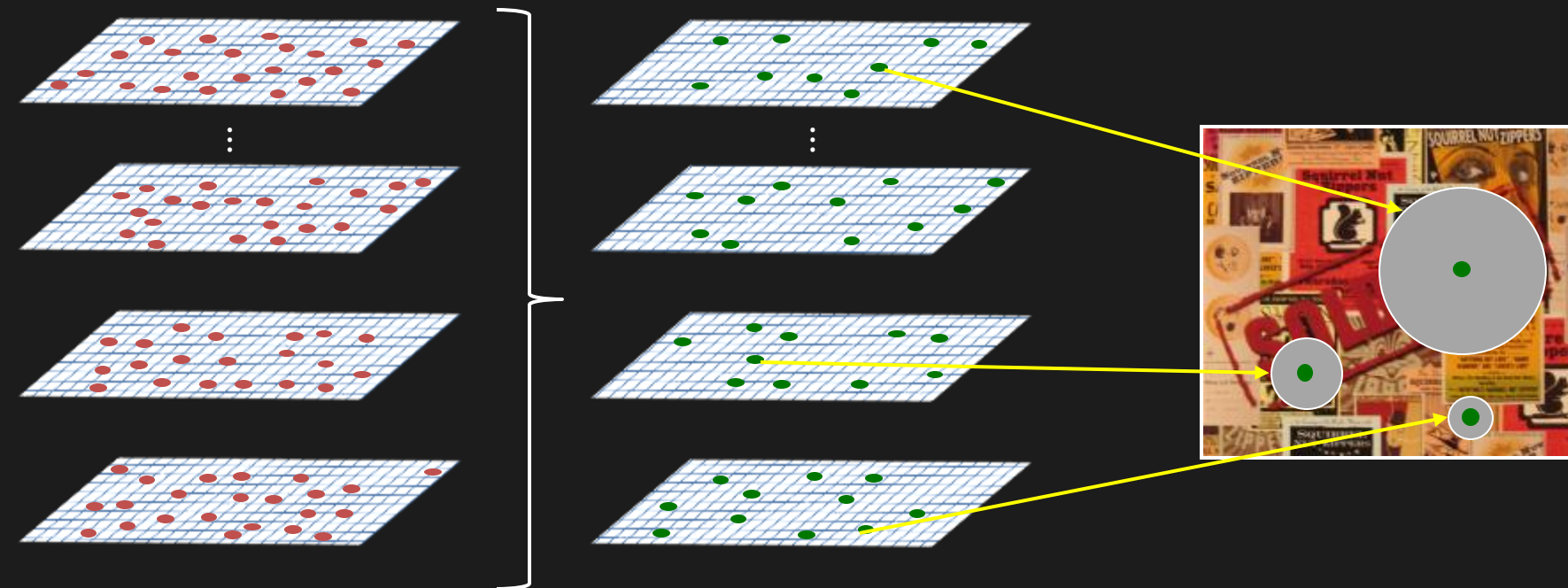
Extracting SIFT Interest Points



Extracting SIFT Interest Points



Extracting SIFT Interest Points

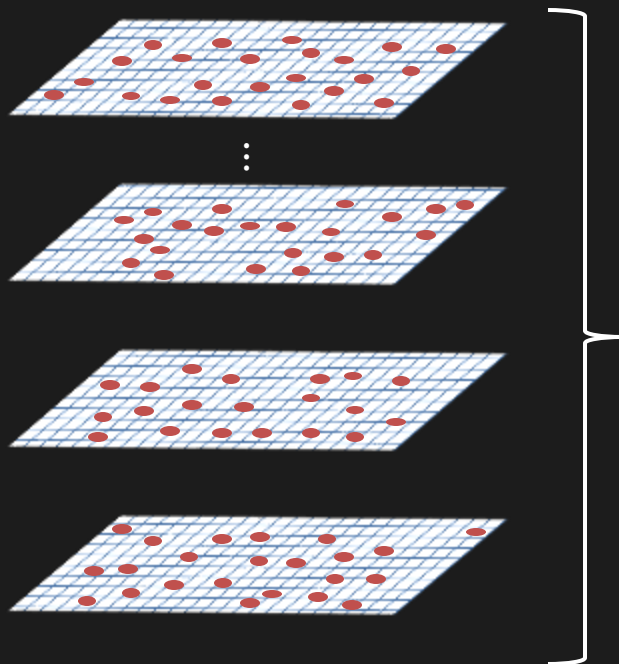


Interest Point
Candidates

(includes weak extrema,
bad contrast, ...)

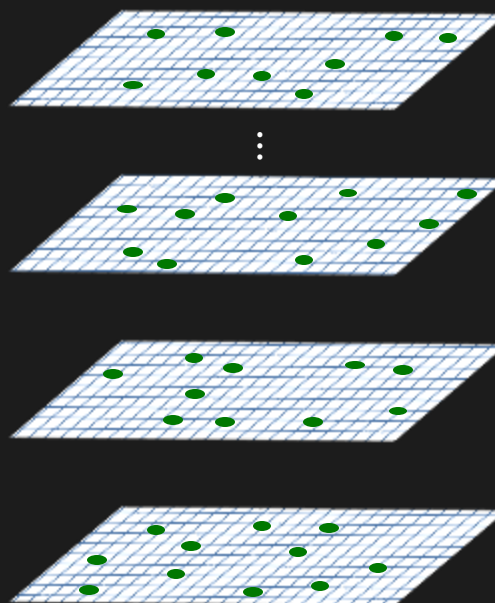
SIFT
Interest Points
(after removing
weak points)

Extracting SIFT Interest Points



Interest Point
Candidates

(includes weak extrema,
bad contrast, ...)

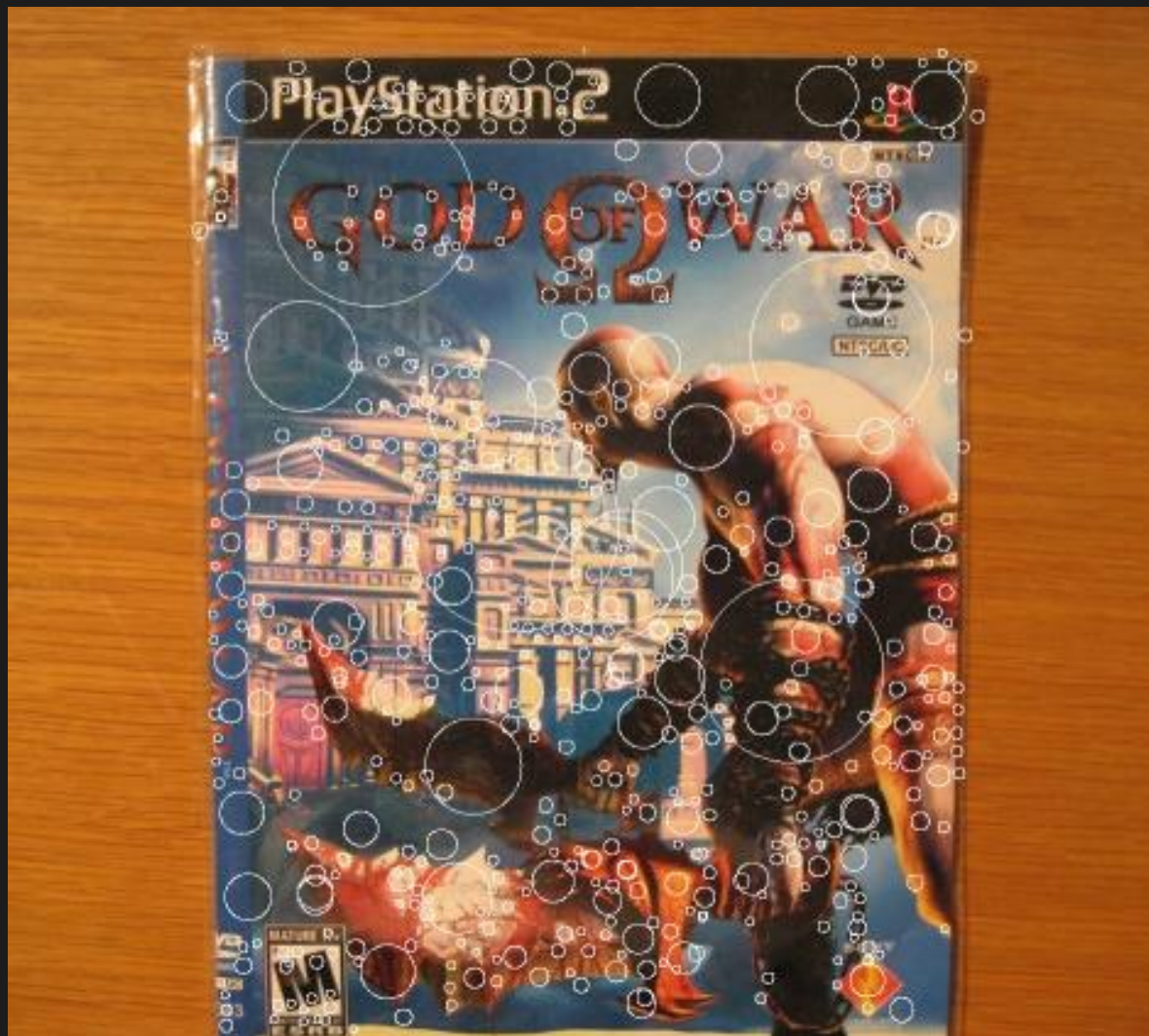


SIFT
Interest Points
(after removing
weak points)



Interest Point
Depiction

SIFT Detection Examples



SIFT Detection Examples



SIFT Detection Examples



2D Blob Detection Summary

Given an image $I(x, y)$.

Convolve the image using NLoG at many scales σ .

Find: {

$$(x^*, y^*, \sigma^*) = \arg \max_{(x, y, \sigma)} |\sigma^2 \nabla^2 n_\sigma * I(x, y)|$$

or

$$(x^*, y^*, \sigma^*) = \arg \max_{(x, y, \sigma)} |\sigma^2 \nabla^2 S(x, y, \sigma)|$$

(x^*, y^*) : Position of the blob

σ^* : Size of the blob

Scale-Space



$S(x, y, \sigma_0)$



$S(x, y, \sigma_1)$



$S(x, y, \sigma_2)$



$S(x, y, \sigma_3)$

...

Increasing σ , Higher Scale, Lower Resolution

Scale Space: Stack created by filtering an image with Gaussians of different sigma (σ)

$$S(x, y, \sigma) = n(x, y, \sigma) * I(x, y)$$

Creating Scale-Space



$S(x, y, \sigma_0)$



$S(x, y, \sigma_1)$



$S(x, y, \sigma_2)$



$S(x, y, \sigma_3)$

...

Increasing σ , Higher Scale, Lower Resolution

Selecting sigma's to generate the scale-space:

$$\sigma_k = \sigma_0 s^k \quad k = 0, 1, 2, 3, \dots$$

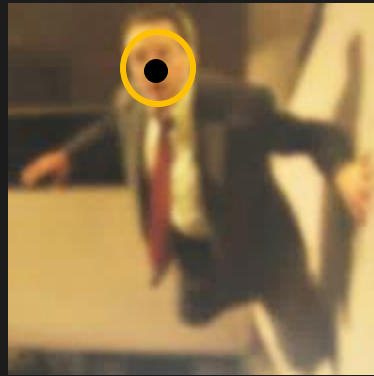
where, s : Constant multiplier

σ_0 : Initial Scale

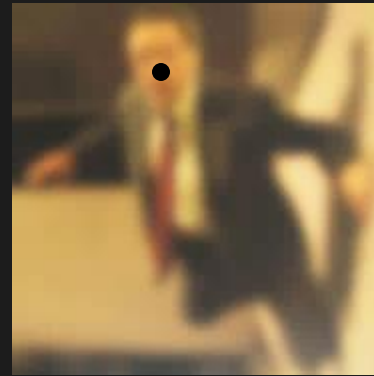
Blob Detection using Local Extrema



$S(x, y, \sigma_0)$



$S(x, y, \sigma_1)$



$S(x, y, \sigma_2)$

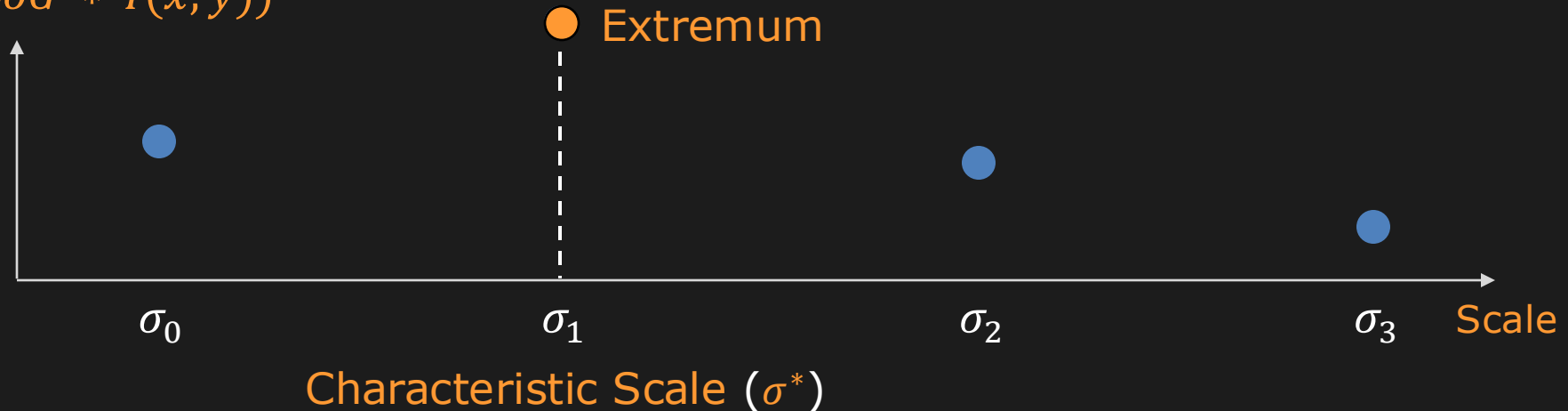


$S(x, y, \sigma_3)$

...

$$\sigma^2 \nabla^2 S(x, y, \sigma)$$

(NLoG * I(x, y))



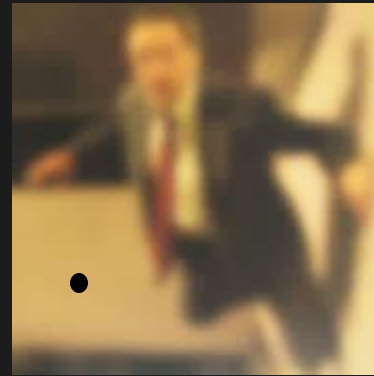
Blob Detection using Local Extrema



$S(x, y, \sigma_0)$



$S(x, y, \sigma_1)$



$S(x, y, \sigma_2)$

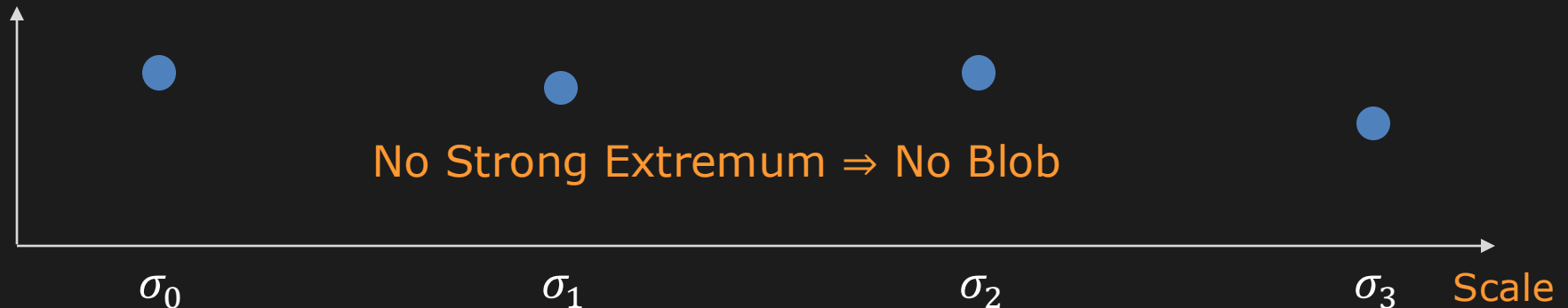


$S(x, y, \sigma_3)$

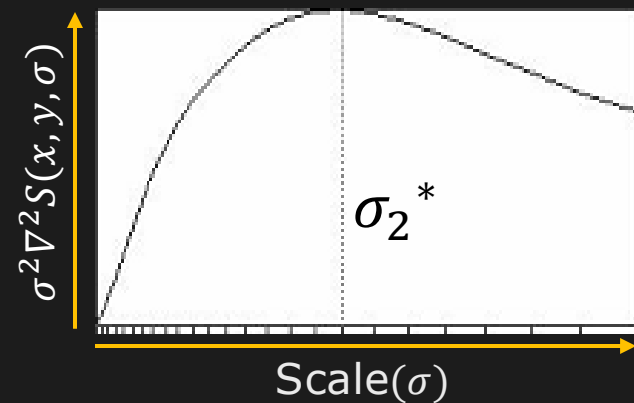
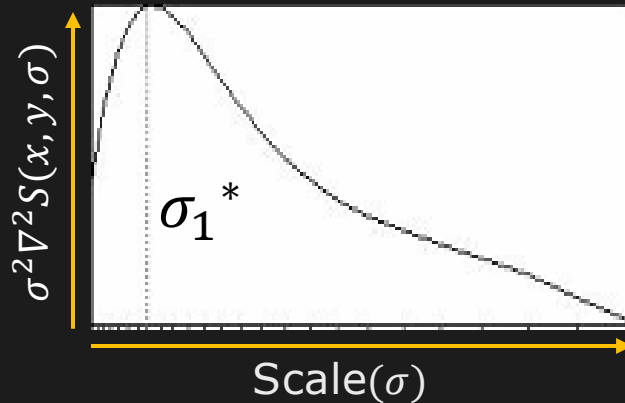
...

$$\sigma^2 \nabla^2 S(x, y, \sigma)$$

(*NLoG* * $I(x, y)$)



Comparison of Characteristic Scales



$$\frac{\sigma_1^*}{\sigma_2^*} : \text{Ratio of Blob Sizes}$$

Computing the Principal Orientation

Use the histogram of gradient directions

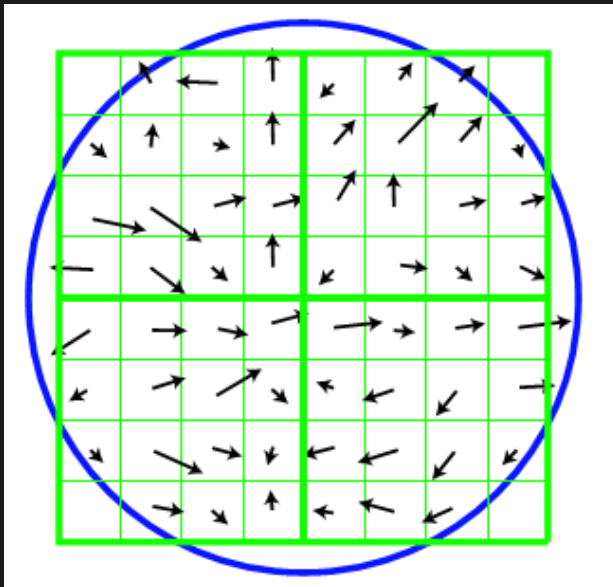
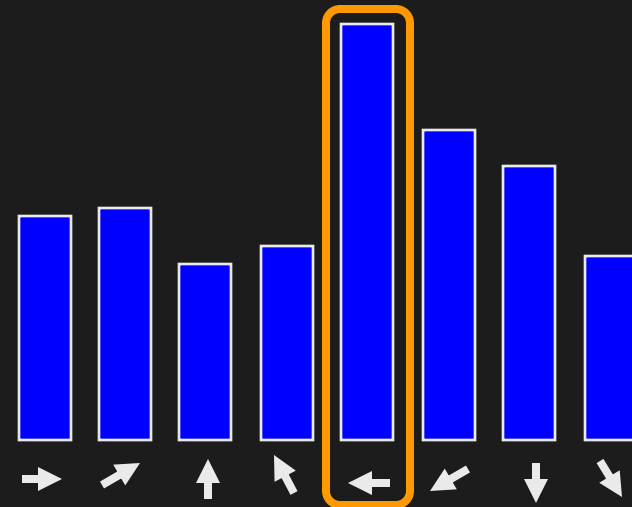


Image gradient directions

$$\theta = \tan^{-1} \left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x} \right)$$

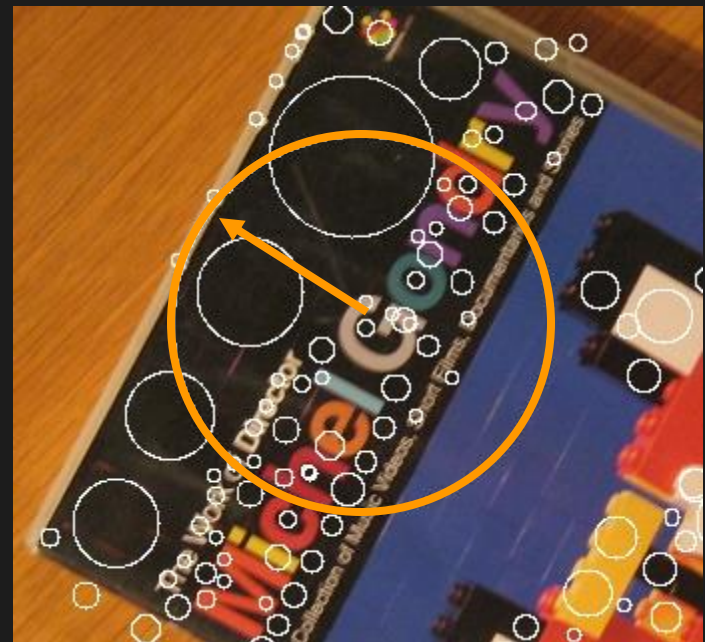
Principal Orientation



Choose the most prominent gradient direction

SIFT Rotation Invariance

Use the principal orientations to match rotation



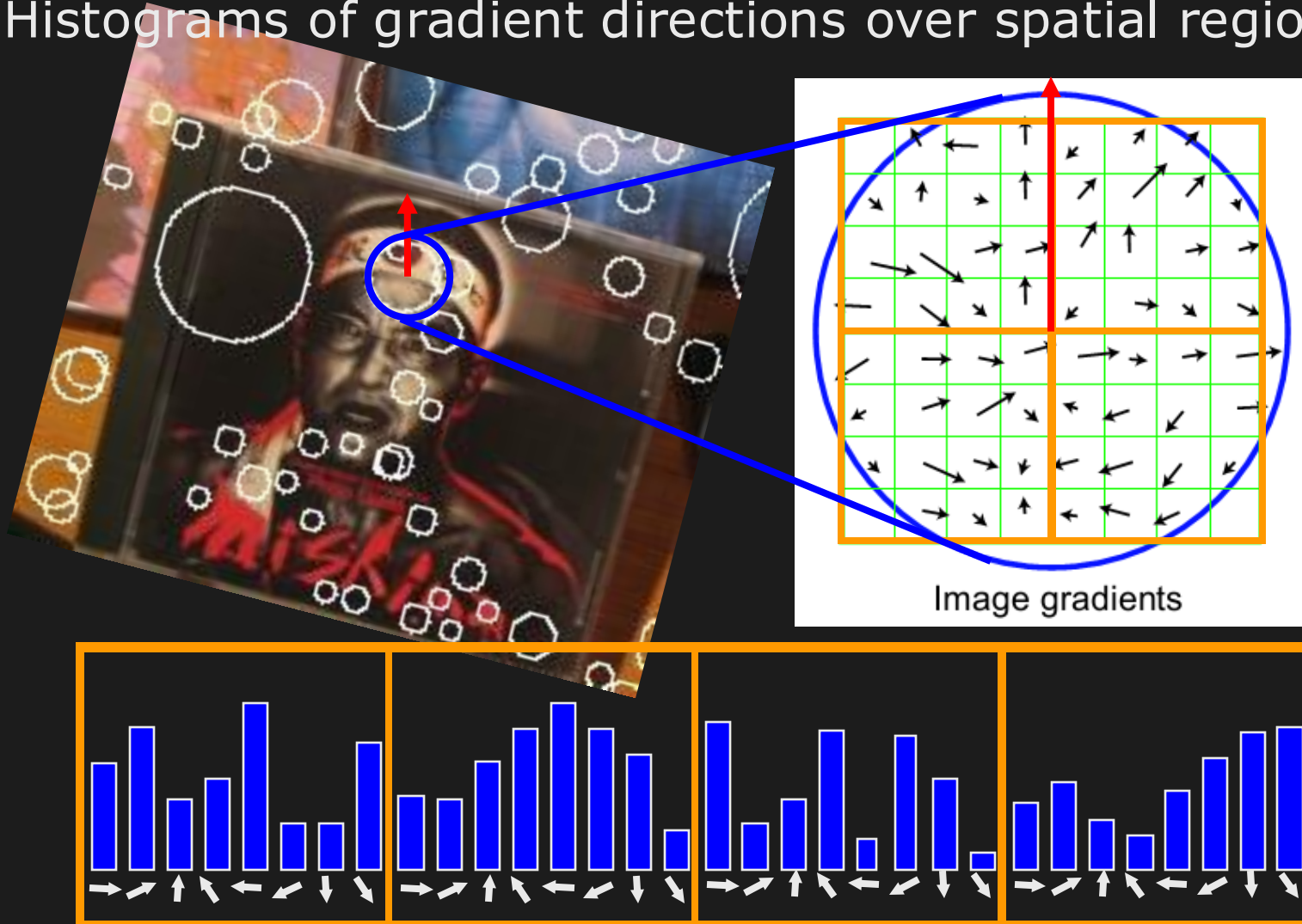
The SIFT Descriptor

Histograms of gradient directions over spatial regions



The SIFT Descriptor

Histograms of gradient directions over spatial regions



Normalized Histogram: Invariant to Rotation, Scale, Brightness

[Lowe 2004]

Comparing SIFT Descriptors

Essentially comparing two arrays of data.

Let $H_1(k)$ and $H_2(k)$ be two arrays of data of length N .

L2 Distance:

$$d(H_1, H_2) = \sqrt{\sum_k (H_1(k) - H_2(k))^2}$$

Smaller the distance metric, better the match.

Perfect match when $d(H_1, H_2) = 0$

Comparing SIFT Descriptors

Essentially comparing two arrays of data.

Let $H_1(k)$ and $H_2(k)$ be two arrays of data of length N .

Correlation:

$$d(H_1, H_2) = \frac{\sum_k [(H_1(k) - \bar{H}_1)(H_2(k) - \bar{H}_2)]}{\sqrt{\sum_k (H_1(k) - \bar{H}_1)^2} \sqrt{\sum_k (H_2(k) - \bar{H}_2)^2}}$$

$$\text{where: } \bar{H}_i = \frac{1}{N} \sum_{k=1}^N H_i(k)$$

Larger the distance metric, better the match.

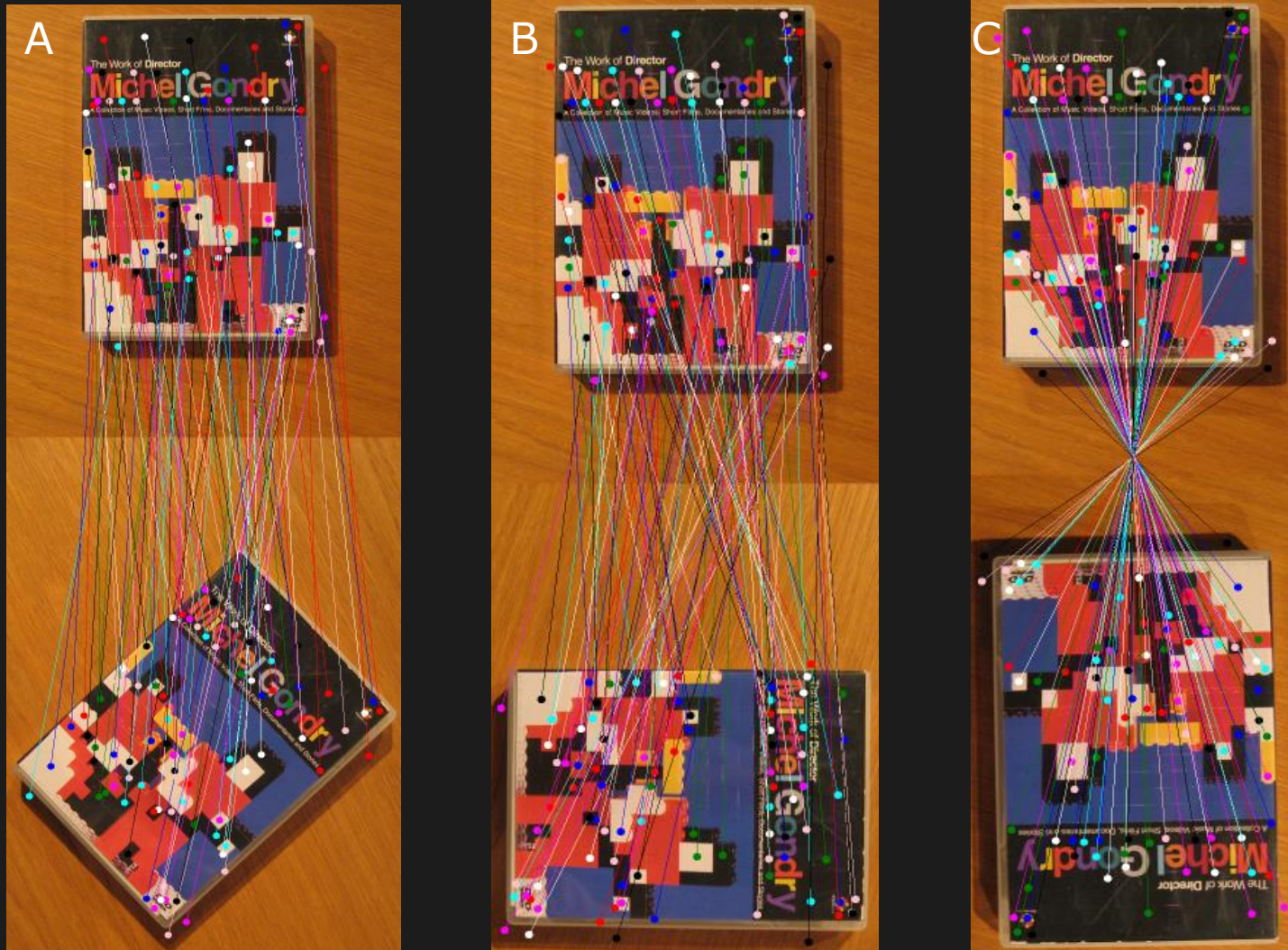
Perfect match when $d(H_1, H_2) = 1$

SIFT Results: Scale Invariance



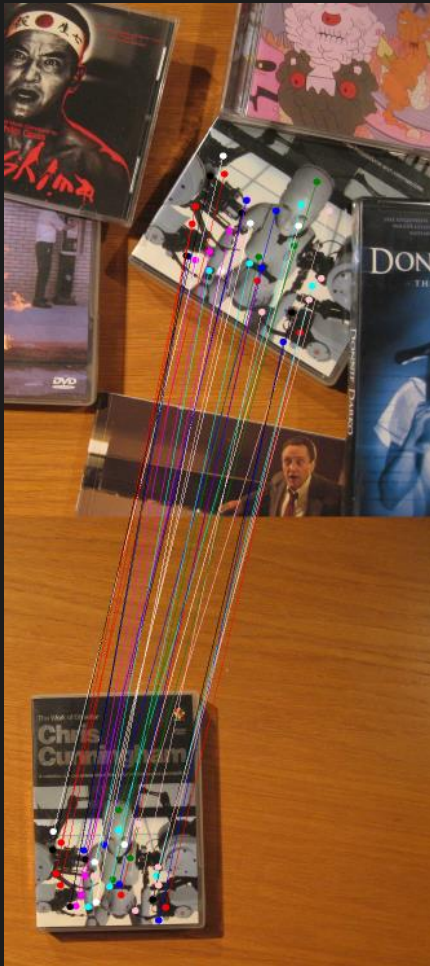
SIFT detects corresponding features in images at different resolutions

SIFT Results: Rotation Invariance



SIFT detects corresponding features in rotated images

SIFT Results: Robustness to Clutter



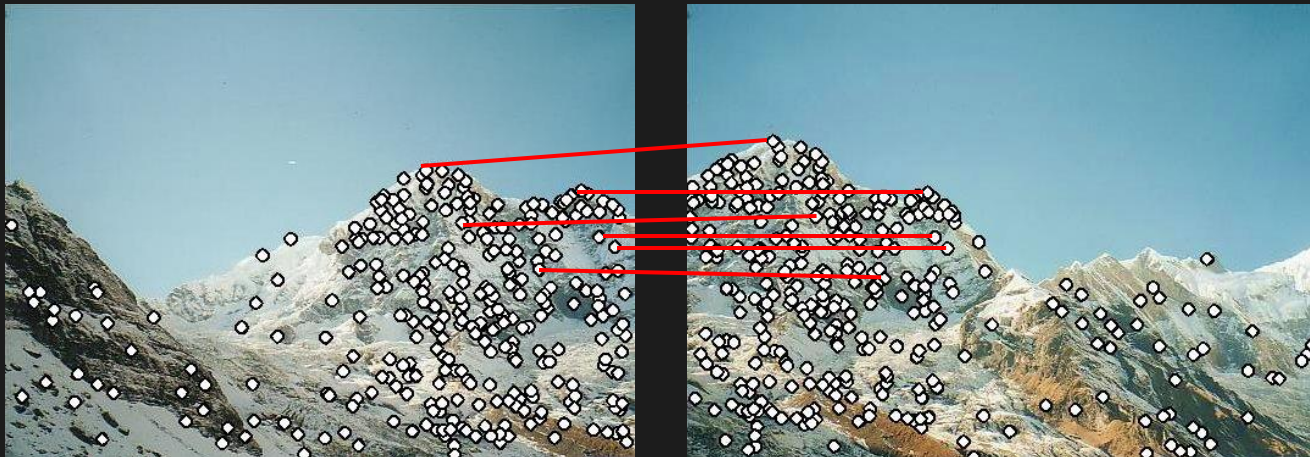
Panorama Stitching using SIFT



Image 1



Image 2



Match SIFT Interest Points

Panorama Stitching using SIFT

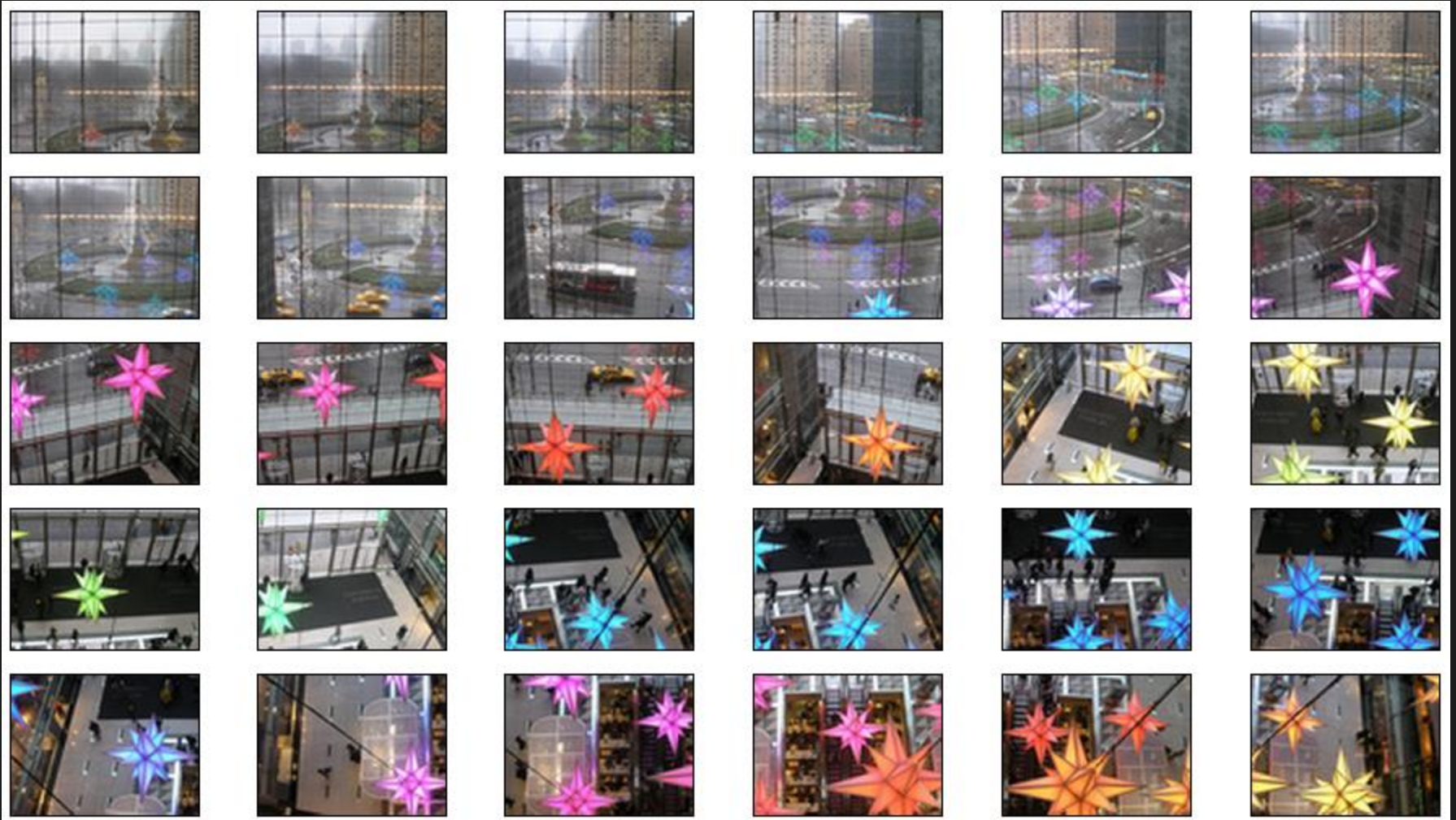


Align and combine images to create a larger image

[More in next lecture]

[Autostitch]

Auto Collage using SIFT



Captured Photos

Auto Collage using SIFT

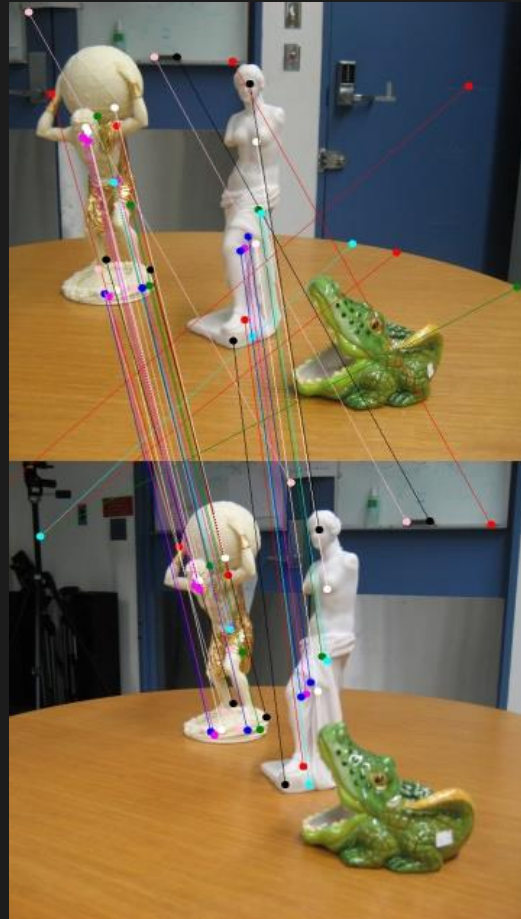
Collage



SIFT for 3D Objects?



No Change in Viewpoint



30° Change in Viewpoint



90° Change in Viewpoint

SIFT is reliable for only small changes in viewpoint

References: Textbooks

Computer Vision: Algorithms and Applications

Szeliski, R., Springer

References: Papers

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[Witkin 1983] A. Witkin. "Scale-Space Filtering". *IJCAI*, 1983.